The Role of Machine Learning in Mammogram Analysis for Early Breast Cancer Detection

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Abstract

At present, Breast cancer is the most seen common cancer and growing widely among ladies all over the world, and early prediction of breast cancer is crucial for successful treatment and improve our outcomes. Among all cancers Breast Cancer covers 20% in new cases. Detection of Breast Cancer is tough for experts also. To detect it accurately many methods and algorithms are developed and implemented by our experts to reduce the death rate.
Mammography is one of the better modalities to detect breast cancer and it decreased death rates due to breast cancer by early detection. We took reference of many good papers, and our main objective is to predict whether the person is suffering with breast cancer in early stage of the cancer and protect people in the early stage.

Keywords: Breast Cancer, Diagnostic Accuracy, Early detection, Mammograms

Introduction

Mammography is a type of medical imaging that uses low-dose x-rays to create images of breast tissue. It is widely used as a screening tool for the detection of breast cancer in asymptomatic women as well as for the diagnosis of breast cancer in symptomatic women.
[1]Mammography works by compressing breast tissue between two plates and then taking X-ray images from different angles. A radiologist checks the resulting images for any abnormalities, such as masses or calcifications, that could indicate breast cancer. In recent years, major advances have been made in mammography technology, including digital mammography and 3D mammography (also known as tomosynthesis), which provide better image quality and more accurate detection breast cancer. Computer-aided detection (CAD) is another tool used in mammography that uses software to help radiologists analyse mammograms and identify potential abnormalities. Although mammography is effective in detecting breast cancer, it has certain limitations. False positives and false negatives can occur, resulting in unnecessary biopsies or missed diagnoses, respectively.

Breast cancer, like other cancers, grows out of control due to the proliferation of cells in a woman's breast.

Most of the Cancer Society and other organisations recommend that women are at risk for breast cancer start getting mammograms at age 40 and continue getting it every 1-2 years so we can keep the information open to everyone and detect and treat cancer early. There are many algorithms to test and many of them have the best accuracy.

Literature review

Table 01: Literature Review

Author and year	Topic	Research design/methodology	Limitations	Key findings
Liao et al. (2022)	The role of supplemental breast ultrasound in mammogram-negative dense breasts	Randomized controlled trial	Single-center study design	Supplemental ultrasound improves detection of breast cancer in females with mammogram-negative thick breasts.
Ahmed et al. (2019)	Use of tomosynthesis (3D mammography) for breast cancer detection	Meta Analysis	Variations in study design and quality	Tomosynthesis has higher sensitivity and specificity for detecting breast cancer, particularly in women with thick breasts.
Zhu et al. (2020)	The effect of COVID-19 on breast cancer Screening and diagnosis	Retrospective analysis	Limited sample size and single-center study design	Breast cancer screening and diagnosis decreased during the pandemic due to reduced access and fear of exposure
Nguyen et al. (2021)	Artificial intelligence (AI) in mammography interpretation	Systematic review	Limited standardization of AI algorithms and lack of head-to-head comparisons	AI can improve the accuracy and efficiency of mammogram interpretation
Lee et al. (2020)	Comparison of mammogram and ultrasound for breast cancer detection	Prospective study	Single-center study design	Ultrasound has higher sensitivity but lower specificity than mammography for detecting breast cancer
Kim et al. (2019)	Use of computer-aided detection (CAD) for mammogram interpretation	Randomized controlled trial	Limited generalizability to other populations and settings	CAD improves sensitivity but reduces specificity in mammogram interpretation
Durand et al. (2020)	Performance on detection of breast cancer in mammography and digital breast tomosynthesis images by machine learning algorithms	Retrospective study	Limited sample size	using mammography to detect the cancer and digital breast tomosynthesis images.

Early detection of cancer is essential for fortunate treatment and improved progress. In recent years, several techniques and methods have been developed to detect breast cancer, including mammograms.
Reviewed more than 15 previous papers and mentioned some of them above. From all these papers, observed process and their limitations. So, we tried to improve their Limitations by increasing sample size for effective accuracy to maintain the model for future use.

Methodology

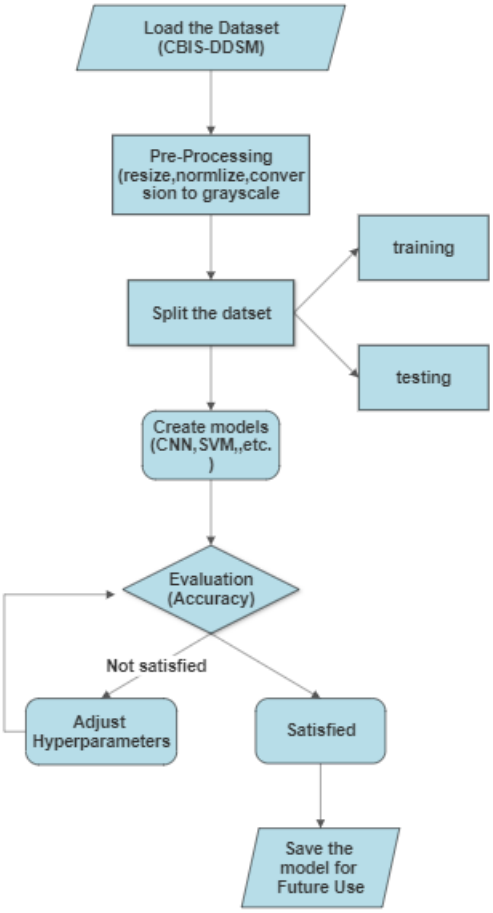

Figure 1: Flow chart for Methodology

Data Used: CBIS-DDSM is a publicly available dataset of mammography images used for breast cancer research. The images path given to breast histopathology images. Here, Number of Images of no cancer: 198738, Number of Images of cancer: 78786, Total Number of Images: 277524 with accompanying labels indicating the presence or absence of breast cancer

Image pre-processing is an important step in many computer vision tasks, including breast cancer detection on mammograms using machine learning algorithms. Here are some common pre-processing techniques that can be applied to mammograms to improve the performance of machine learning algorithms:

Data Cleaning: this involves removing unwanted data, Changing datatypes, columns name for better understanding and analysis. Here, cropped images are 3859, ROI mask images are 3340, full mammogram images are 3038.then examined the breast parts from the collected images the analysis in the following way Total Breast are 9671, Left Breast are 318, Right Breast are 248.Furthermore, found the two abnormalities and their count in the following way

calcification are 1546, mass is 1318, in the images. left or right breast and their count in the way, Right are 728, Left are 818 and made some necessary changes.

Noise Reduction: Medical images can be affected by noise or artifacts that interfere with image interpretation. Techniques such as filtering or smoothing can be used to reduce noise and improve image quality.
Image normalization: [2]This involves adjusting the intensity levels of an image to make it more consistent and easier to analyse. For mammograms, this might involve rescaling pixel values to a specific range, such as 0 to 255, or applying a logarithmic transformation to compress the dynamic range of the image.

Image enhancement: This involves applying filters or other transformations to improve the contrast, sharpness, or other visual characteristics of an image. For mammograms, this may involve applying a median filter to remove noise or using adaptive histogram equalization to improve the visibility of fine structures.

Feature Extraction: This involves the identification and quantification of specific visual features of images relevant to the task at hand. For mammograms, this may involve measuring the size, shape and texture of suspicious lesions or extracting radiomic features from images.

Data Augment**:** This involves creating additional training examples by applying random transformations to the original images, such as rotating, flipping**,** or scaling**.**

In general, in this case it helped a lot to remove null values. The pre-treatment plays a important role in this detecting breast cancer before by improving the quality and accuracy of medical images, making it easier for us and healthcare professionals to identify early signs of breast cancer.

Programming Language Used

Used Python Language because, in this case of early detection of breast cancer, Python can be used to analyze large amounts of data from medical imaging, such as mammograms and ultrasounds. Python has a wide range of powerful libraries and frameworks. it can be used to develop image processing and analysis.

Libraries used: TensorFlow, NumPy, keras, Matplotlib, Pandas, sickit-learn, CV2, PyTorch

CNN (Convolutional Neural Network):

Neural networks are often used in image processing, natural language processing, and other tasks involving data processing. CNNs are particularly useful for tasks such as image classification, object detection, and segmentation.
CNNs have multiple layers, including communication, integration, and full-scale layers. Convolutional layers are responsible for analysing and extracting the characteristics of the input data, while clustering layers help reduce the dimensionality of the data and make it more traceable. All layers are used to classify the input data based on the extracted values.
Here, the 2D convolution layer has 32 filters, the filter size is 3x3, the padding is set to "the same", and the ReLU activation function takes an input of size 50x50x3. A 2D max pooling layer with step 2.
Generally, this code describes a CNN model with multiple interlinked layers and layered layers combined with layered layers for classification. Overall, CNNs have proven useful for many tasks and are widely used in science and industry.

Result Generation

Accuracy: Accuracy is a key metric used to evaluate the machine learning algorithms performance. It measures the Percentage of correctly classified instances among all instances in the dataset.

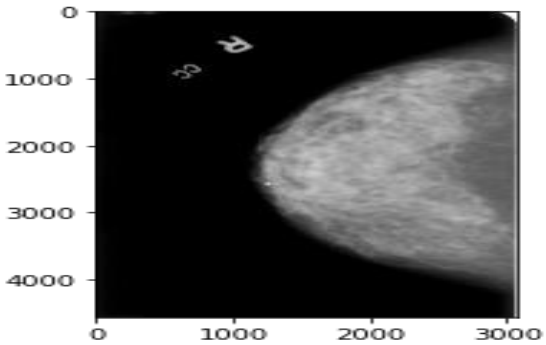

Figure 01: ROI image on Gray scale

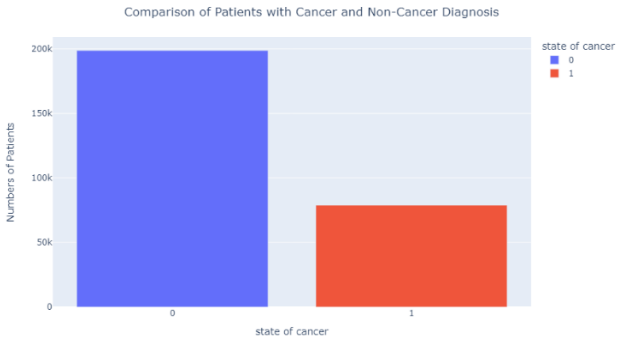

Figure 02:Patients vs cancer state

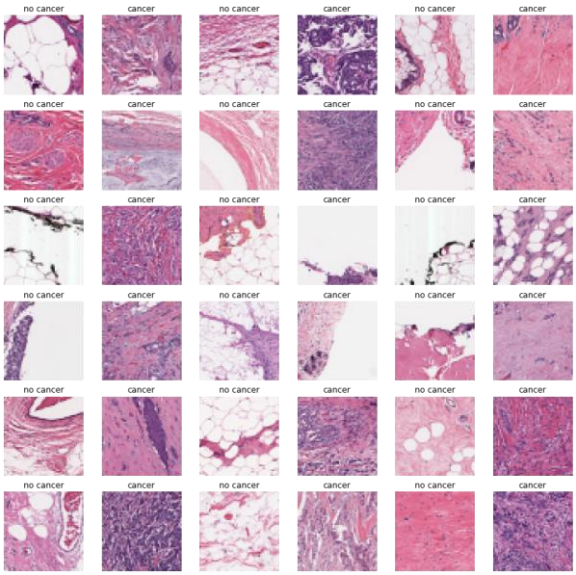

Figure 03: Images of cancer and no cancer

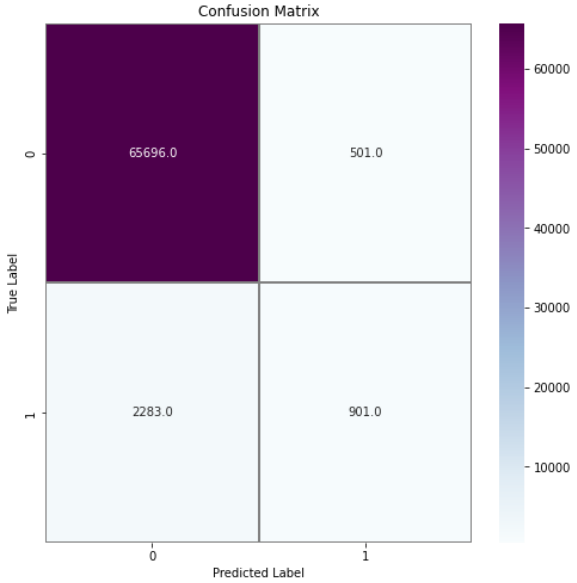

Figure 04: Confusion Matrix

Figure 05: Model Accuracy

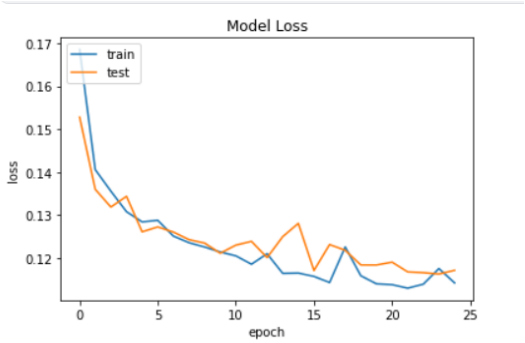

Figure 06: Model Loss

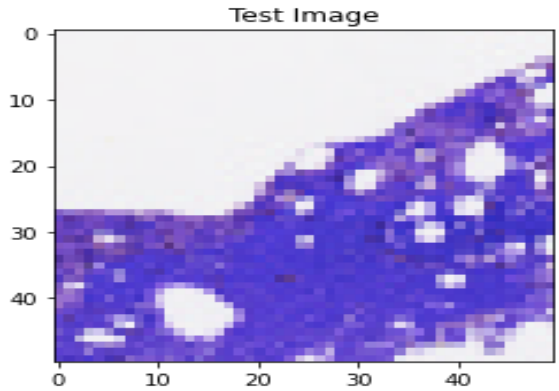

Figure 07: Test image

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Predicted Value using  cnn model 0
True Value 0
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Figure 08: Output for Test Image

Conclusion

Mammography is an important tool for early detection of breast cancer, and its use has led to significant improvements in breast cancer mortality rates. However, mammography is not perfect, and false-positive results and missed cancers are still a concern. Newer technologies such as DBT and breast MRI can help to address these limitations and improve the accuracy of breast cancer detection.

It is important for women to discuss breast cancer screening options with their health care providers to determine which approach best suits their individual needs.

Therefore, the study recommends that government agencies and other organizations help the areas where there are no proper mammography tools with expertise on use. The study also recommends further research to uncover the relationship between different screening methods and for age groups to make clear the problem of false positives and false negatives. The study also suggest research on knowledge and use of mammography in rural developing countries.

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